

Moving around

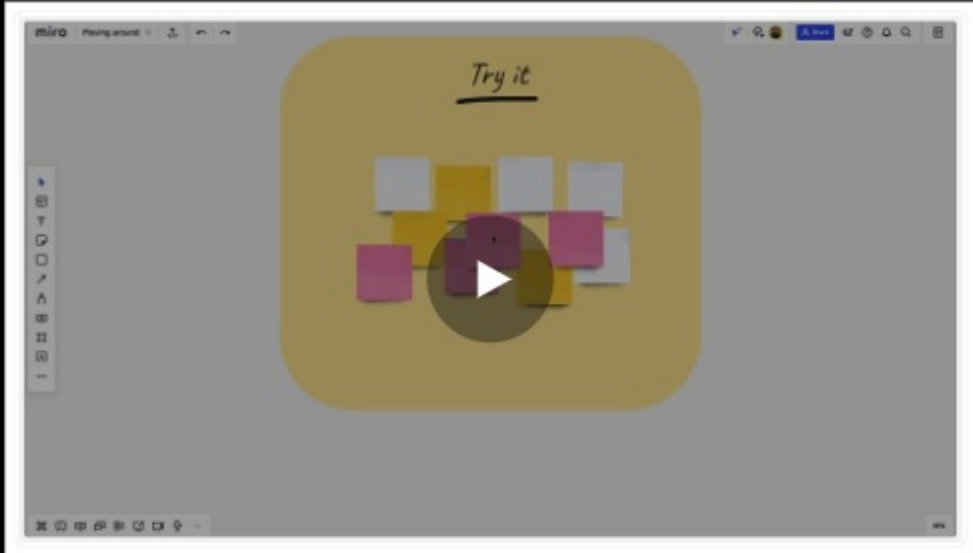
Hold spacebar then click and drag to move around.

Find it



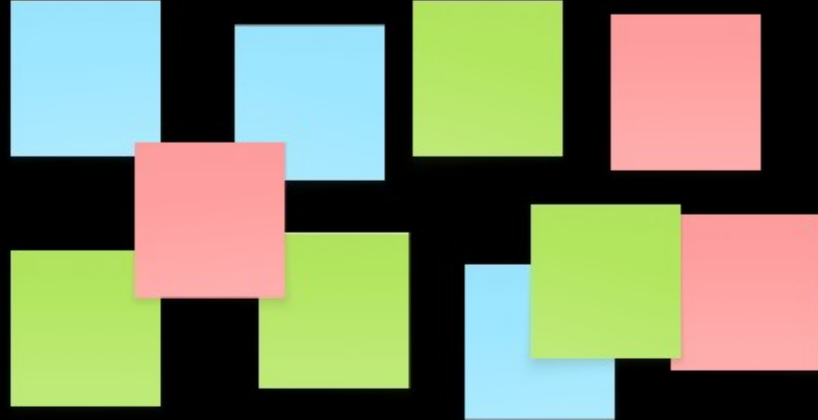
Hold **spacebar** then click and drag to move around

See it



Try it

- ① Hold **spacebar** until you see a hand
- ② Place your **cursor** over the stickies, then **click and drag**



 Edit the board when your cursor looks like a **pointer** (v). Move around with the **hand** (h).

Zooming in and out

Use the plus and minus keys on your keyboard to zoom in and out.

Find it



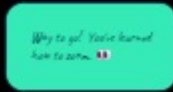
Use the **plus** and **minus** keys on your keyboard

See it



Try it

- ① Center your screen on the small text below
- ② Use the **plus** and **minus** keys to zoom in and read the message

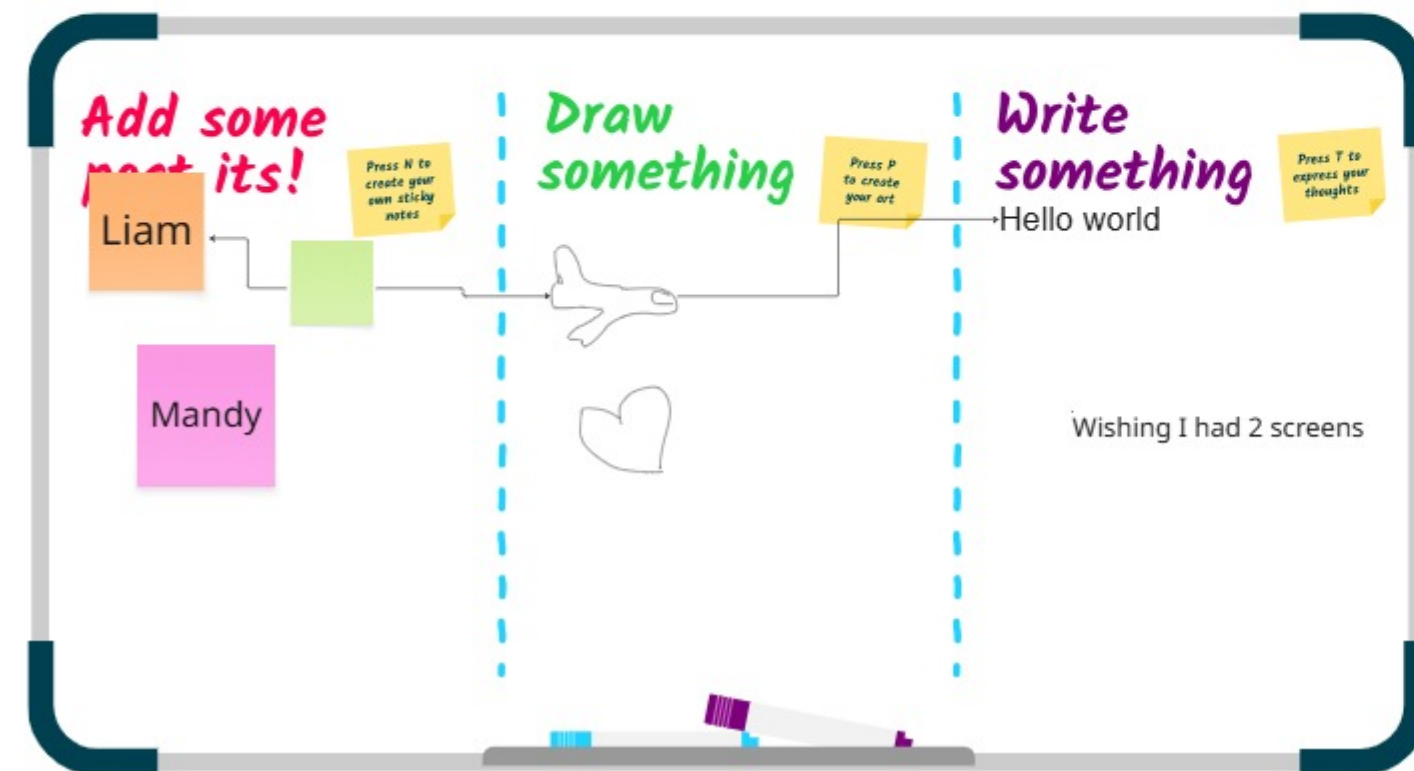


 Alternatively, use the **scroll wheel** on a mouse or **pinch** on a trackpad.

Have a play

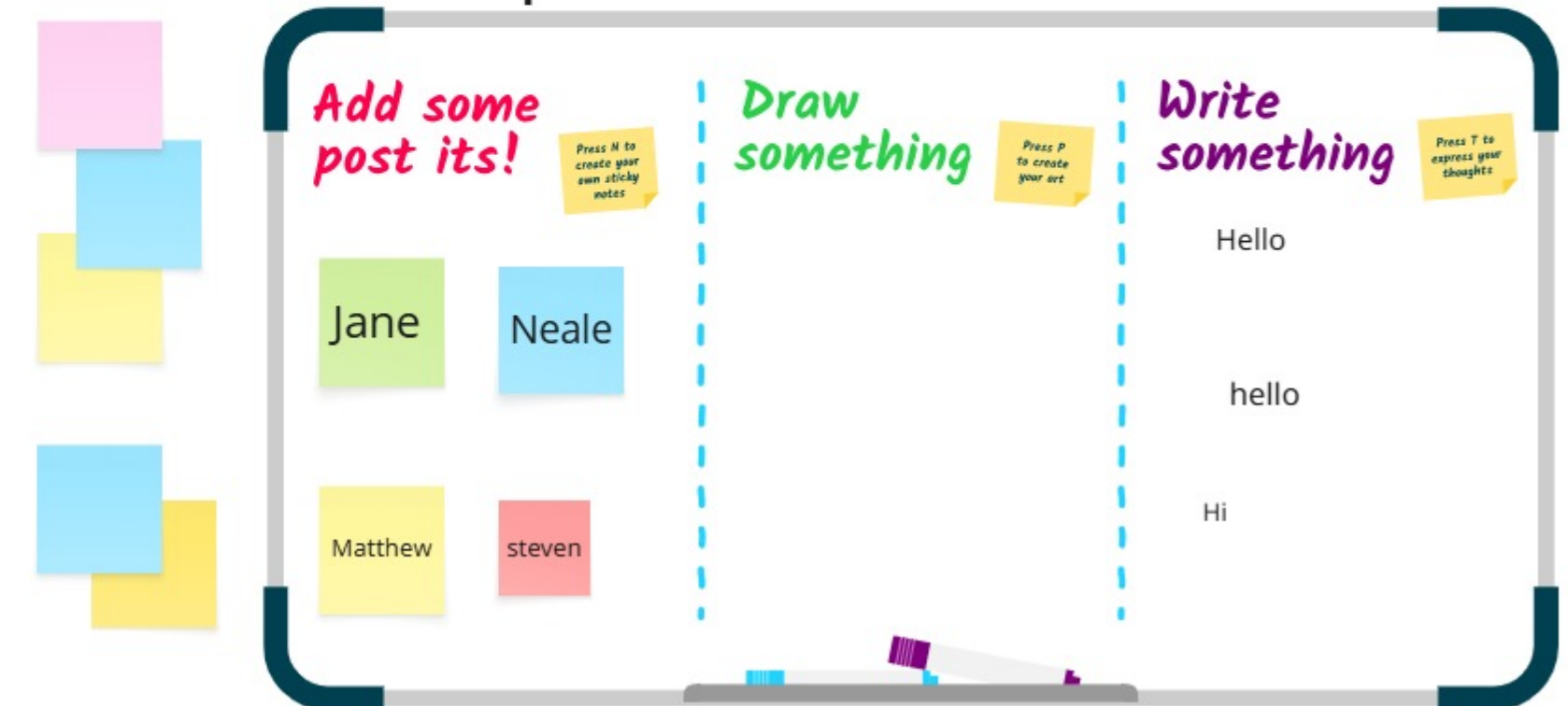
Use the tools to get familiar with adding content.

Group 1



Group 1 digital whiteboard with three columns: 'Add some post its!', 'Draw something', and 'Write something'. The 'Add some post its!' column contains sticky notes for 'Liam' and 'Mandy'. The 'Draw something' column features a drawing of a bird and a heart. The 'Write something' column has the text 'Hello world' and 'Wishing I had 2 screens'. A toolbar at the bottom shows drawing tools like a brush and eraser.

Group 2



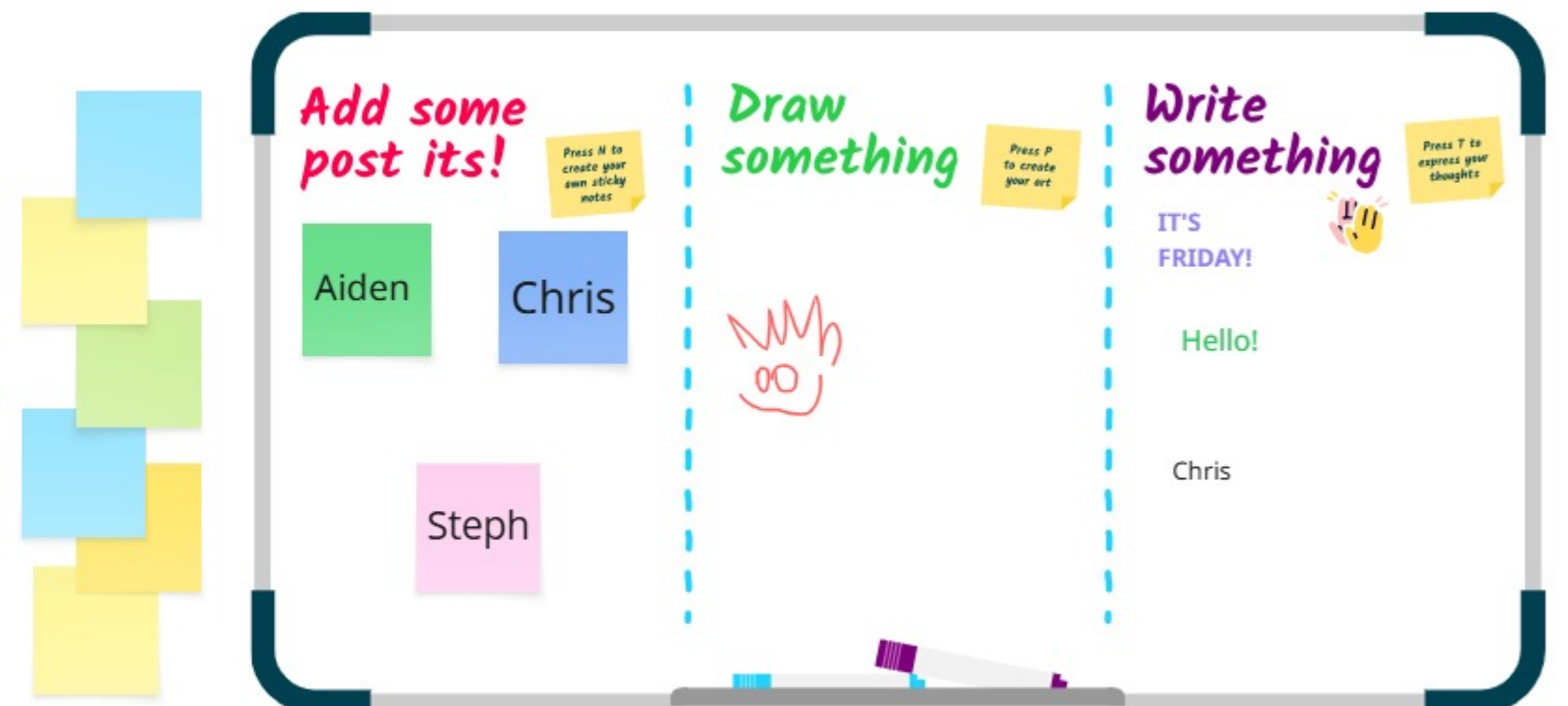
Group 2 digital whiteboard with three columns: 'Add some post its!', 'Draw something', and 'Write something'. The 'Add some post its!' column contains sticky notes for 'Jane', 'Neale', 'Matthew', and 'steven'. The 'Write something' column has the text 'Hello', 'hello', and 'Hi'. A toolbar at the bottom shows drawing tools like a brush and eraser.

Group 3



Group 3 digital whiteboard with three columns: 'Add some post its!', 'Draw something', and 'Write something'. The 'Add some post its!' column contains sticky notes for 'Ewan', 'Blair', 'Kurt', and 'Nneka'. The 'Draw something' column features a drawing of a smiley face, a box, and a cat. The 'Write something' column has the text 'Friday!', 'Hello :)', and 'Good Morning'. A toolbar at the bottom shows drawing tools like a brush and eraser.

Group 4



Group 4 digital whiteboard with three columns: 'Add some post its!', 'Draw something', and 'Write something'. The 'Add some post its!' column contains sticky notes for 'Aiden', 'Chris', and 'Steph'. The 'Draw something' column features a drawing of a hand. The 'Write something' column has the text 'IT'S FRIDAY!', 'Hello!', and 'Chris'. A toolbar at the bottom shows drawing tools like a brush and eraser.

Housekeeping

Rules regarding webcams

Keep your Webcam on throughout the delivery

Rules regarding mute

Keep your Microphone off throughout for privacy unless you are asked or need to respond.

Frequency of breaks

Short Breaks are scheduled every hour to support you with your learning.

Rules regarding group participation

Everyone should equally participate in Group discussion and take lead in contributing your view.

Rules regarding questions

The Golden Rule:

'There is no such thing as a stupid question'

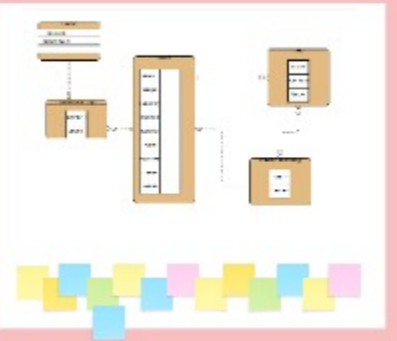
'Even when asked by an instructor'

Please have a go at answering questions

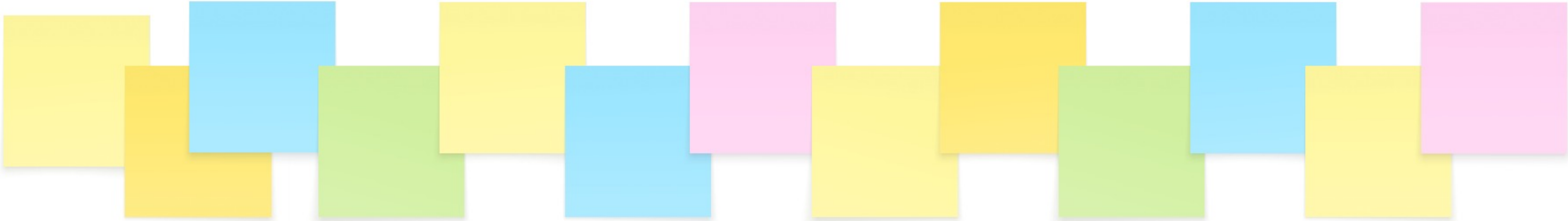
Also:

'A question never resides in a single mind'
By asking questions, you are helping everyone out!

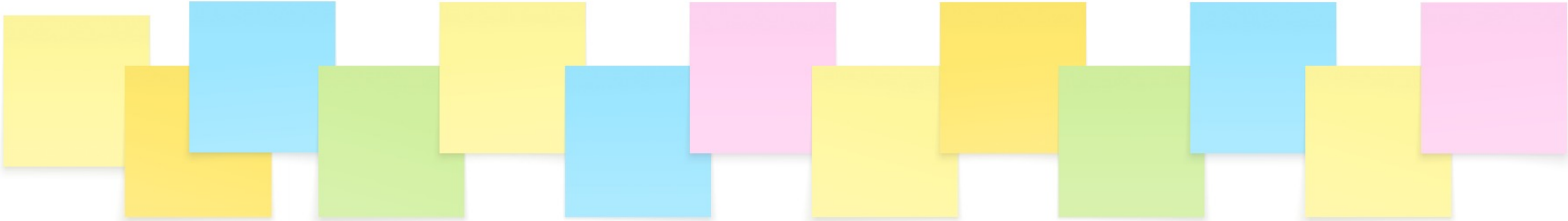
Anything else



	Column	Column1	Column 2	Column 3	+
1	Entity description				
2	Database				
3	Entity name	Movies			
4	Key	Attributes	Data Types	Constraints/comments	
5	PK	MoviesID	Int	NotNull	
6					
7					
8					
+					



	Column	Column1	Column 2	Column 3	+		
1	Entity description						
2	Database						
3	Entity name						
4	Key	Attributes	Data Types	Constraints/comments			
5							
6							
7							
8							
+							



Teams

	Fran	Liam	Mandy									
	Jane	Matthew	Neale	Steven								
	Blair	Ewan	Kurt	Nneka								
	Aiden	Chris	Stephanie									

Go to your team board


Group 1


Group 2


Group 3

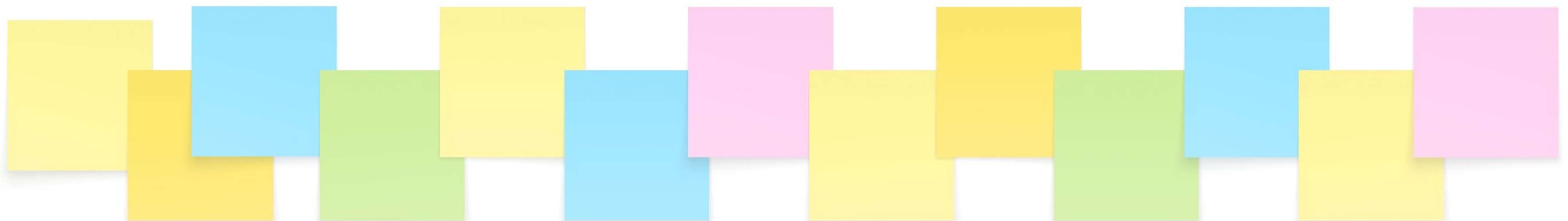

Group 4

Task 1

Identify and name potential entities from the information supplied.

There are at least 5 entities needed and 2 of these are bridging tables.

1. DirectorID
2. MovieID
3. DirectorID/Movie Bridge (M:M)
4. ActorID
5. Character/Actor Bridge (M:M)

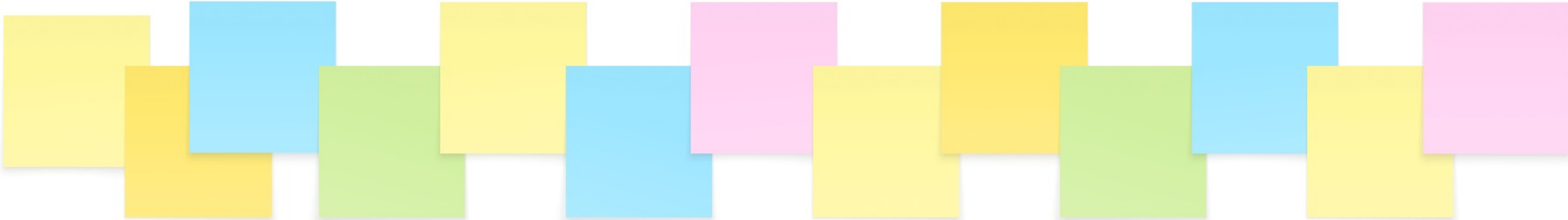


Movies and Actors Database

Director	Actor	gender	character_name	cast_order	movie_id	title	budget	homepage	overview	popularity	release_date	revenue	runtime	movie_status	tagline	vote_average	vote_count
Quentin Tarantino	Tim Roth	Male	Ted the Bellhop	43	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hands	8	651
Robert Rodriguez	Tim Roth	Male	Ted the Bellhop	44	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hands	8	651
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Alexandre Rockwell	Tim Roth	Male	Ted the Bellhop	46	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hands	8	651
Quentin Tarantino	Antonio Banderas	Male	Man	47	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hands	8	651
Robert Rodriguez	Antonio Banderas	Male	Man	48	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hands	8	651
Allison Anders	Antonio Banderas	Male	Man	49	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hands	8	651
Alexandre Rockwell	Antonio Banderas	Male	Man	50	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hands	8	651
Quentin Tarantino	Jennifer Beals	Female	Angela	51	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hands	8	651
Robert Rodriguez	Jennifer Beals	Female	Angela	52	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hands	8	651
Allison Anders	Jennifer Beals	Female	Angela	53	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hands	8	651
				54	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hands	8	651

The attributes in the Movies and Actors table are:

- Director
- Actor
- gender
- character_name
- cast_order
- movie_id
- title
- budget
- homepage
- overview
- popularity
- release_date
- revenue
- runtime
- movie_status
- tagline
- vote_average
- vote_count



Column

Column1

Column 2

Column 3

Entity description

Database

Entity name

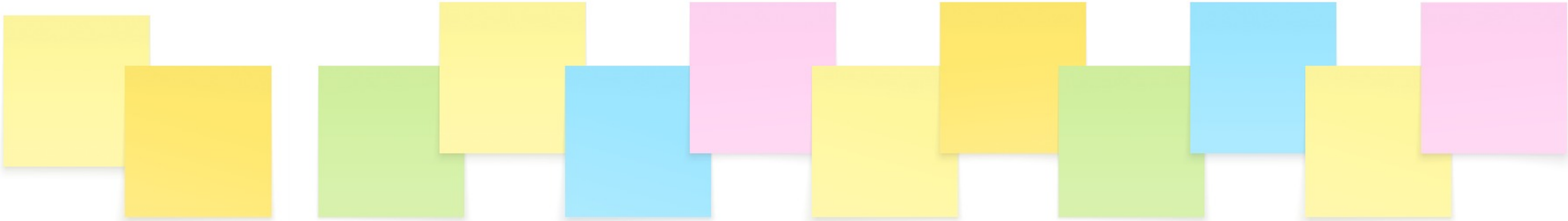
Key

Attributes

Data Types

Constraints/comments

Diagram



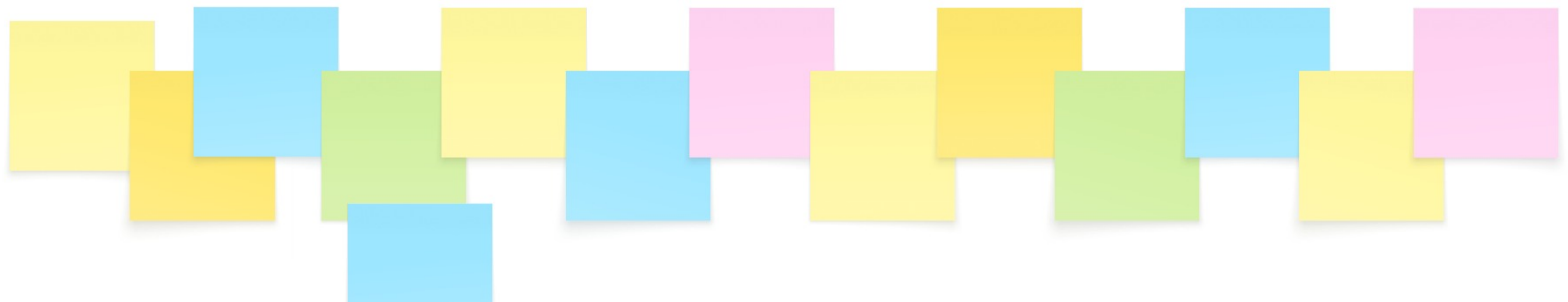
Director	
DirectorID	
DirectorName	

Director/Movie Bridge	
DirectorID	
MovieID	

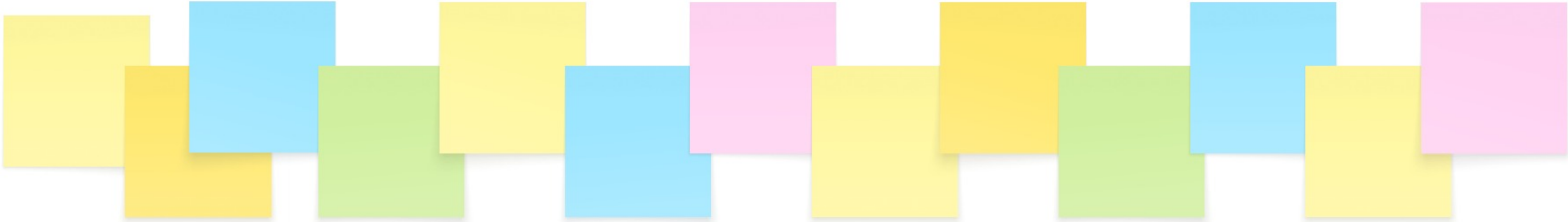
Movies	
MovieID	
Budget	
Popularity	
ReleaseDate	
overview	
Tagline	
MovieStatus	
Runtime	
Character	

Actor		
ActorID		
ActorName		
Gender		

CharacterID/Title Bridge	
ActorID	
CharacterID	



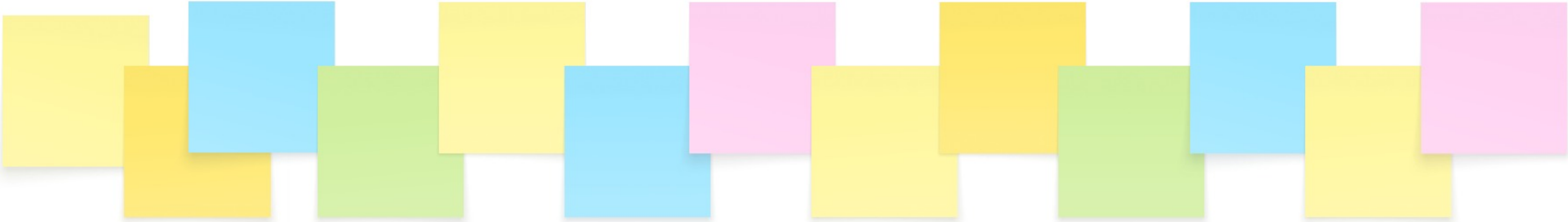
	Column	Column1	Column 2	Column 3	+		
1	Entity description						
2	Database						
3	Entity name						
4	Key	Attributes	Data Types	Constraints/comments			
5							
6							
7							
8							
+							



Task 2

Compile data dictionaries for your identified entities.
You may use the pro forma on the following page (or any other valid format).
You should include any Keys (Primary and Foreign), Data Types for each attribute and any suitable constraints I.e. Null, Not Null

	Column	Column1	Column 2	Column 3	+
1	Entity description	Director for the Movies			
2	Database	Movies			
3	Entity name	Director			
4	Key	Attributes	Data Types	Constraints/comments	
5	PK	Director ID	Int	Not Null	
6		Director name	Vchar(50)	Not Null	
7					
8					
+					



Activity 1: Movies Database Normalisation

Learner instructions

We have been given the task of creating an online resource based on movies produced in the last 50 years.

One of the features of this resource will be a RDMS that will store details of the personnel and production information relating to a large number of films.

We have an Excel file provided with data that has been partly normalised in that some of the entities and their attributes have already been identified.

However one of the sheets on the Excel file labelled Movies and Actors still needs to be normalised before it is suitable for the RDMS.

Screenshots can be seen below of this table and its attributes.

Task 1

The attributes in the Movies and Actors table are:

Director

Actor

gender

character_name

cast_order

movie_id

title

budget

homepage

overview

popularity

release_date

revenue

runtime

movie_status

tagline

vote_average

vote_count

Identify and name potential entities from the information supplied.

Remember that there may be many-to-many relationships between the tables that would require a bridging table (a table containing the foreign keys of the tables to be joined)

Think about the fact that and actor can play many characters but these characters can appear in a number of movies i.e. Luke Skywalker appears in Star Wars but also in The Empire Strikes Back!

Also when you create any of the main tables from they will need a Primary Key attribute that may need to be added for example director_ID

Identify and name potential entities from the information supplied.

Hint:

There are at least 5 entities needed and 2 of these are bridging tables.

a.

a.

b.

c.

d.

Task 2

Compile data dictionaries for your identified entities.

You may use the pro forma on the following page (or any other valid format).

You should include any Keys (Primary and Foreign), Data Types for each attribute and any suitable constraints I.e. Null, Not Null

Entity description

Database

Entity name

Key

Attribute

Data Type

Constraints/comments

Task 3

Identify entity relationships. Draw an **Entity Relationship Diagram** showing the relationships (cardinality) between the entities(tables) you identified in Task 1.

Show the Primary and Foreign keys to show how tables would relate to each other i.e. The Primary key of one table (Parent) would become the Foreign key in its related table (Child)

Hint:

Bridging tables that are used to avoid many-to-many relationships are basically formed of two one-to-many relationships.

The diagram below shows the “many-to-many” relationship to avoid duplicate values or multiple values in the one column, between Orders and Products.

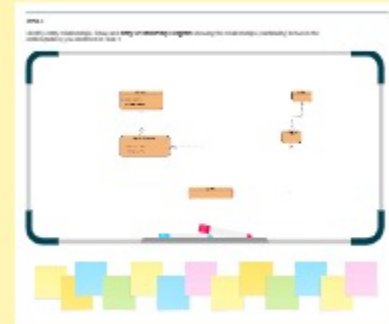
OrderID and ProductID have become foreign keys in the OrderDetails table.

Replace the image above with your own Entity Relationship Diagram (ERD)

<https://online.visual-paradigm.com/>

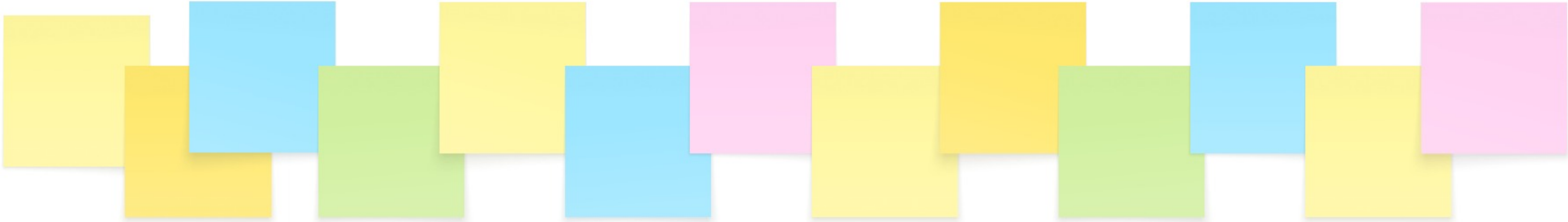


Exercise



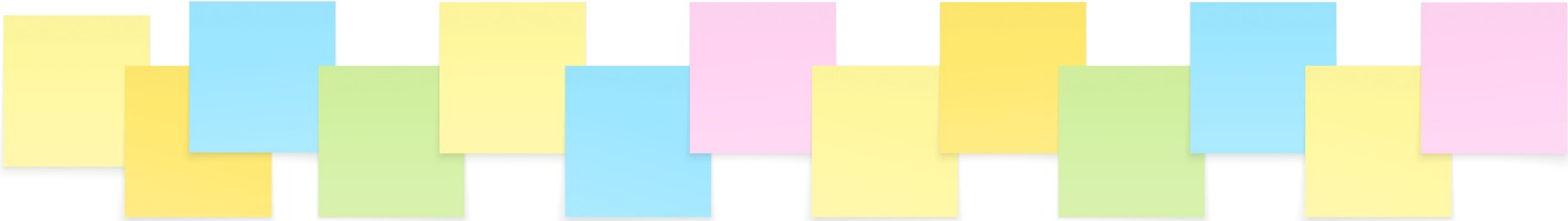
Entity 2

	Column	Column1	Column 2	Column 3	+
1	Entity description	Director Details			
2	Database	Movies			
3	Entity name	Director			
4	Key	Attributes	Data Types	Constraints/comments	
5	PK	Director ID	int	Not Null	
6		Director Name	varchar (50)	Not Null	
7					
8					
+					



Entity 3

	Column	Column1	Column 2	Column 3	
1	Entity description	Movie Details			
2	Database	Movies			
3	Entity name	Movie			
4	Key	Attributes	Data Types	Constraints/comments	
5	PK	Movie ID	int	Not Null	
6		Movie Title	varchar (100)	Not Null	
7					
8		Release Date	date	Not Null	
9		Budget	money		
10		Homepage	varchar(150)		
11		Overview	varchar(max)		
12		Popularity	decimal		
13		Revenue	money		
14		Runtime	int		
15		Tagline	varchar(100)		
16		Vote Average	decimal		
17		Vote Count	Int		
18		Movie Status	varchar(100)		
19		Cast Order	int		
+					



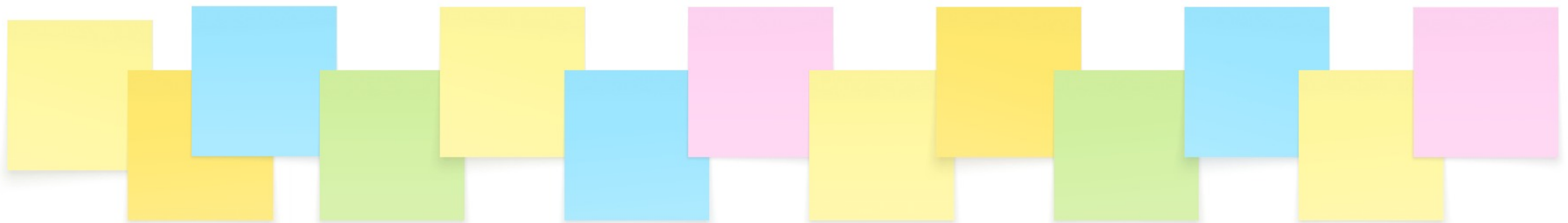
Task 1

Identify and name potential entities from the information supplied.

There are at least 5 entities needed and 2 of these are bridging tables.

1. Director
2. Actor
3. Character (Bridge)
4. Movie
5.Movie Director (Bridge)

t

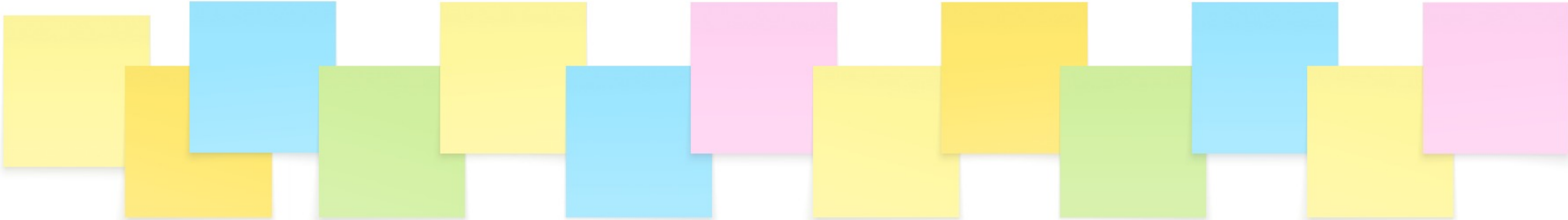


Movies and Actors Database

Director	Actor	gender	character_name	cast_order	movie_id	title	budget	homepage	overview	popularity	release_date	revenue	runtime	movie_status	tagline	vote_average	vote_count
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Robert Rodriguez	Tim Roth	Male	Ted the Bellhop	44	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hands	8	651
Allison Anders	Tim Roth	Male	Ted the Bellhop	45	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hands	8	651
Alexandre Rockwell	Tim Roth	Male	Ted the Bellhop	46	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hands	8	651
Quentin Tarantino	Antonio Banderas	Male	Man	47	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hands	8	651
Robert Rodriguez	Antonio Banderas	Male	Man	48	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hands	8	651
Allison Anders	Antonio Banderas	Male	Man	49	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hands	8	651
Alexandre Rockwell	Antonio Banderas	Male	Man	50	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hands	8	651
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Robert Rodriguez	Jennifer Beals	Female	Angela	52	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hands	8	651
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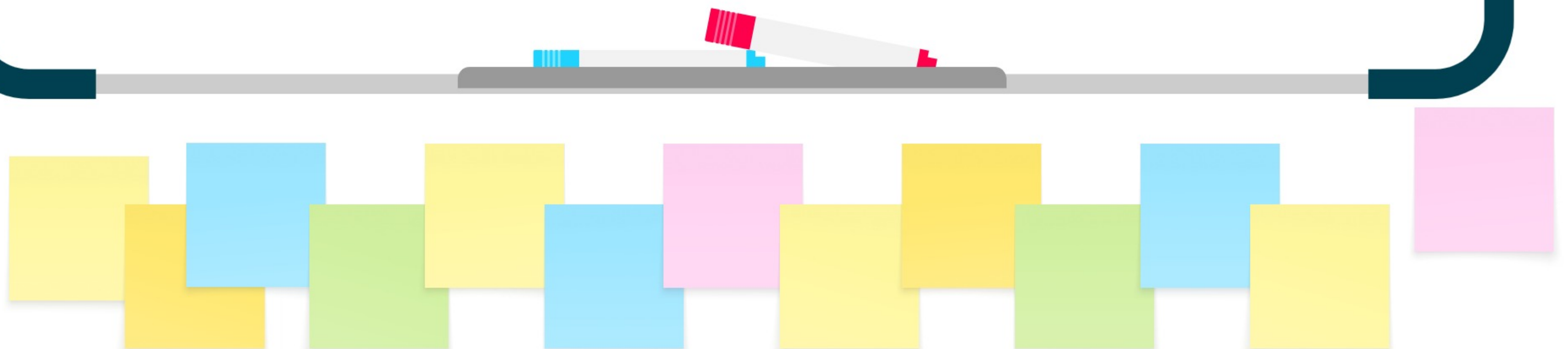
The attributes in the Movies and Actors table are:

- Director
- Actor
- gender
- character_name
- cast_order
- movie_id
- title
- budget
- homepage
- overview
- popularity
- release_date
- revenue
- runtime
- movie_status
- tagline
- vote_average
- vote_count



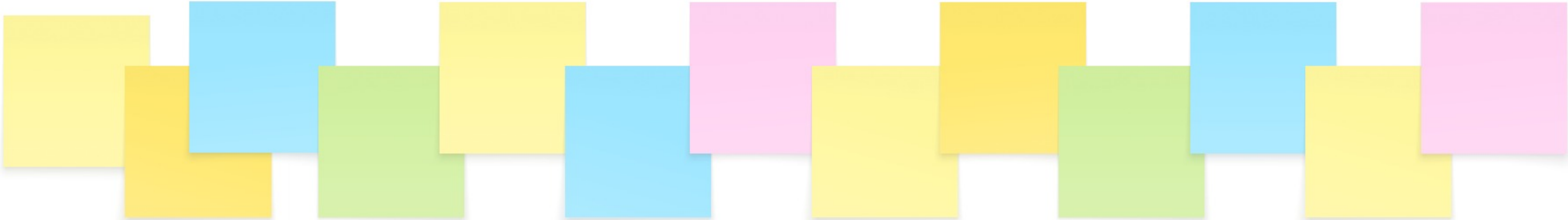
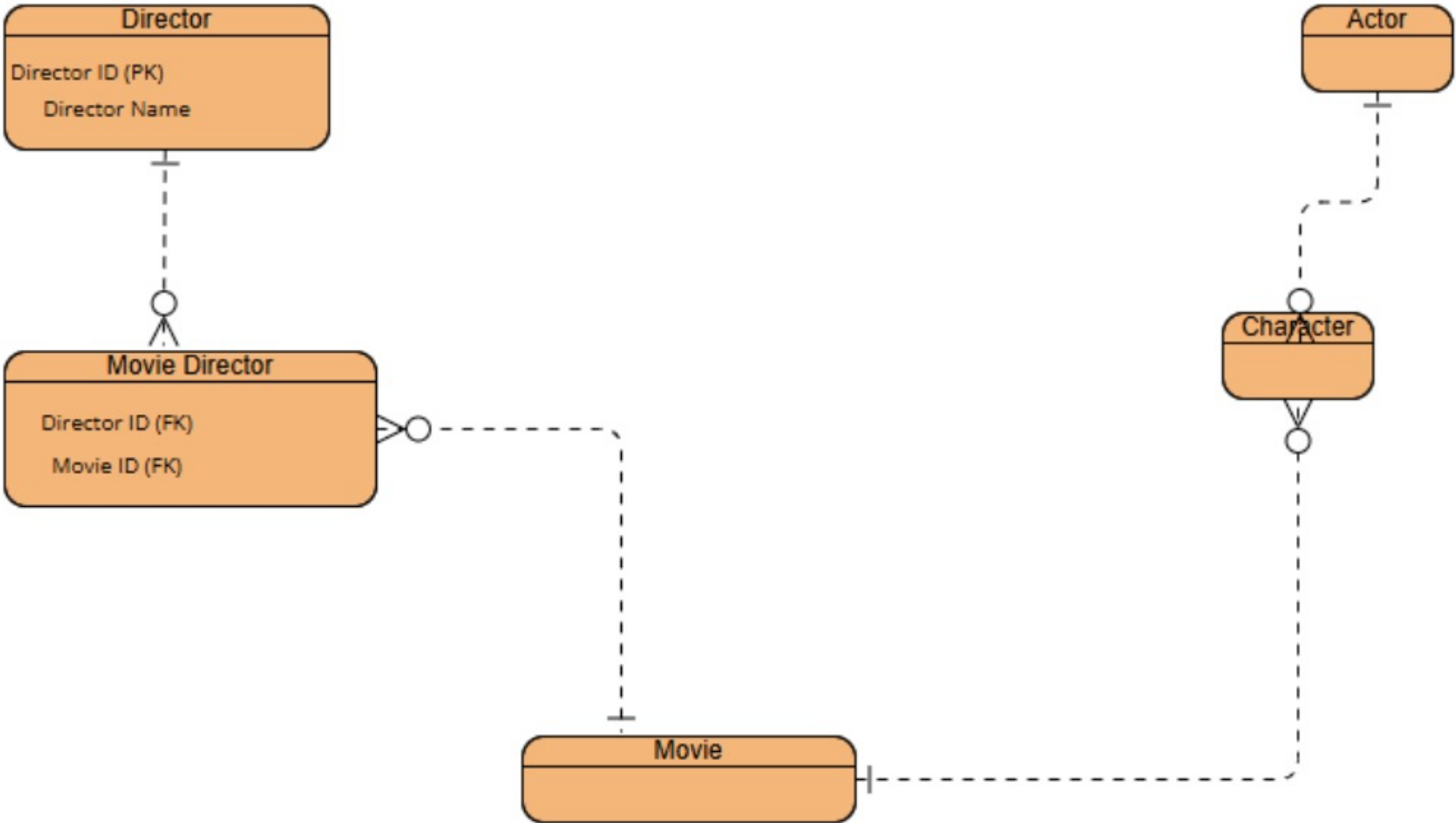
Entity 5

	Column	Column1	Column 2	Column 3	+
1	Entity description	Directors to movies			
2	Database	Movies			
3	Entity name	Movie Diector			
4	Key	Attributes	Data Types	Constraints/comments	
5	FK	Movie ID	int	Not Null	
6	FK	Director ID	int	Not Null	
7					
8					
+					



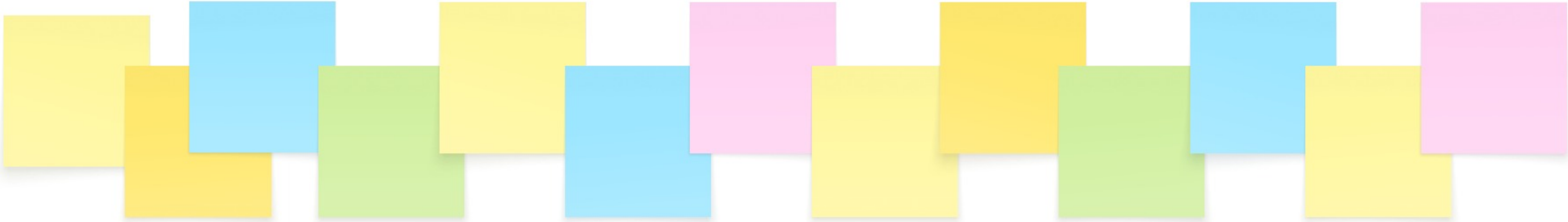
Task 3

Identify entity relationships. Draw an **Entity Relationship Diagram** showing the relationships (cardinality) between the entities (tables) you identified in Task 1.



Entity 4

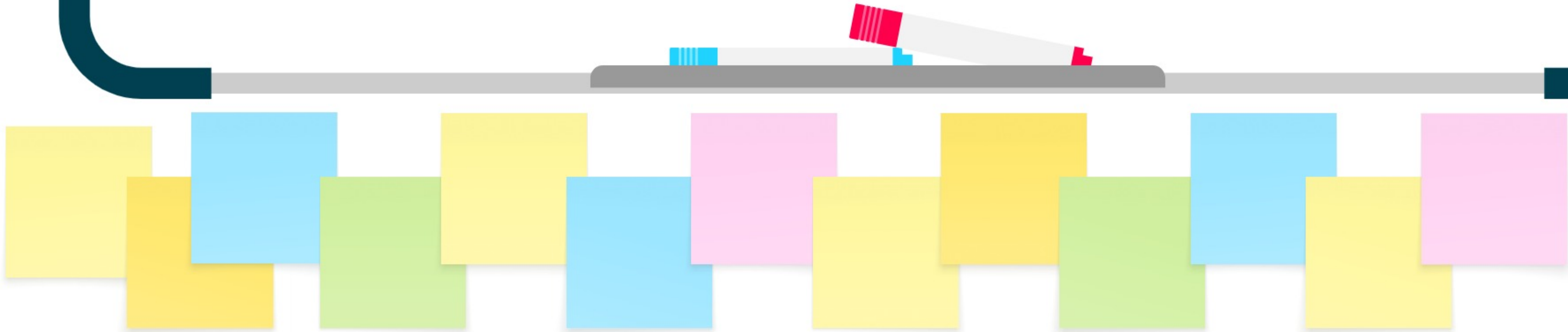
	Column	Column1	Column 2	Column 3	+
1	Entity description	Character Details			
2	Database	Movies			
3	Entity name	Character			
4	Key	Attributes	Data Types	Constraints/comments	
5	FK	Movie ID	int	not null	
6	FK	Actor ID	int	not null	
7		Character Name	varchar (50)	not null	
8					
+					



Task 2

Compile data dictionaries for your identified entities.
You may use the pro forma on the following page (or any other valid format).
You should include any Keys (Primary and Foreign), Data Types for each attribute and any suitable constraints I.e. Null, Not Null

	Column	Column1	Column 2	Column 3	+
1	Entity description	Actor Details			
2	Database	Movies			
3	Entity name	Actor			
4	Key	Attributes	Data Types	Constraints/comments	
5	PK	Actor ID	int	Not Null	
6		Actor Name	varchar (50)	Not Null	
7		Gender	bit	Not Null	
8					
+					





Exercise

Scenario: You are a product manager for a new software product. You have been tasked with identifying the key features and requirements for the product. You have gathered a list of potential features and requirements, and you need to prioritize them based on their importance and feasibility. You have also identified a list of potential users and their needs. Your goal is to create a prioritized list of features and requirements that can be used to guide the development of the product.

Task 1: Identify the key features and requirements for the product. Use the list of potential features and requirements provided below to create a prioritized list of features and requirements. Consider the importance and feasibility of each feature and requirement, and rank them accordingly. The list of potential features and requirements is as follows:

- Feature 1: A user interface that is easy to use and navigate.
- Feature 2: A secure login system that protects user data.
- Feature 3: A reporting system that provides insights into user behavior.
- Feature 4: A mobile app that allows users to access the product on the go.
- Feature 5: A social sharing feature that allows users to share their progress with friends.
- Requirement 1: The product must be able to handle a large volume of users.
- Requirement 2: The product must be able to integrate with existing systems.
- Requirement 3: The product must be able to provide real-time data updates.
- Requirement 4: The product must be able to handle complex data sets.
- Requirement 5: The product must be able to provide a high level of security.

Task 2: Identify the key users and their needs. Use the list of potential users provided below to create a prioritized list of users and their needs. Consider the importance and feasibility of each user and their needs, and rank them accordingly. The list of potential users and their needs is as follows:

- User 1: A small business owner who needs a simple, easy-to-use system to manage their inventory.
- User 2: A large corporation that needs a complex system to manage their supply chain.
- User 3: A government agency that needs a secure system to manage sensitive data.
- User 4: A non-profit organization that needs a system to manage their fundraising efforts.
- User 5: A university that needs a system to manage their student records.

Task 3: Create a prioritized list of features and requirements. Use the list of potential features and requirements provided in Task 1 to create a prioritized list of features and requirements. Consider the importance and feasibility of each feature and requirement, and rank them accordingly. The prioritized list of features and requirements is as follows:

Feature/Requirement	Importance	Feasibility	Rank
Feature 1: A user interface that is easy to use and navigate.	High	High	1
Feature 2: A secure login system that protects user data.	High	High	2
Feature 3: A reporting system that provides insights into user behavior.	Medium	Medium	3
Feature 4: A mobile app that allows users to access the product on the go.	Medium	Medium	4
Feature 5: A social sharing feature that allows users to share their progress with friends.	Low	Low	5
Requirement 1: The product must be able to handle a large volume of users.	High	High	6
Requirement 2: The product must be able to integrate with existing systems.	Medium	Medium	7
Requirement 3: The product must be able to provide real-time data updates.	Medium	Medium	8
Requirement 4: The product must be able to handle complex data sets.	Medium	Medium	9
Requirement 5: The product must be able to provide a high level of security.	High	High	10

Task 4: Create a user persona. Use the list of potential users provided in Task 2 to create a user persona. Consider the needs and goals of each user, and create a persona that represents a typical user of the product. The user persona is as follows:

Persona 1: Small Business Owner

- Age: 35-45
- Gender: Male
- Occupation: Small Business Owner
- Needs: A simple, easy-to-use system to manage their inventory.
- Goals: To increase sales and reduce costs.

Task 5: Create a user journey map. Use the list of potential users provided in Task 2 to create a user journey map. Consider the needs and goals of each user, and create a map that shows the user's journey from initial contact to final purchase. The user journey map is as follows:

Journey Map 1: Small Business Owner

- Initial Contact: The user visits the product website.
- Discovery: The user explores the product features and requirements.
- Consideration: The user compares the product to other options.
- Decision: The user decides to purchase the product.
- Onboarding: The user sets up the product and starts using it.
- Usage: The user uses the product to manage their inventory.
- Support: The user contacts support if they have any issues.
- Feedback: The user provides feedback on their experience.

Task 6: Create a product roadmap. Use the list of potential features and requirements provided in Task 1 to create a product roadmap. Consider the importance and feasibility of each feature and requirement, and create a roadmap that shows the product's development timeline. The product roadmap is as follows:

Product Roadmap 1: Small Business Owner

- Phase 1: Initial development (0-3 months).
- Phase 2: Feature development (3-6 months).
- Phase 3: Testing and deployment (6-9 months).
- Phase 4: Post-launch support (9-12 months).

Task 7: Create a marketing plan. Use the list of potential users provided in Task 2 to create a marketing plan. Consider the needs and goals of each user, and create a plan that shows how to reach and engage with the target audience. The marketing plan is as follows:

Marketing Plan 1: Small Business Owner

- Target Audience: Small business owners.
- Channels: Social media, email, direct mail.
- Messages: Highlight the product's benefits and ease of use.
- Offers: Provide a discount or free trial to encourage sign-ups.
- Timeline: Launch the product in 3 months.

Task 8: Create a business plan. Use the list of potential users provided in Task 2 to create a business plan. Consider the needs and goals of each user, and create a plan that shows how to generate revenue and sustain the business. The business plan is as follows:

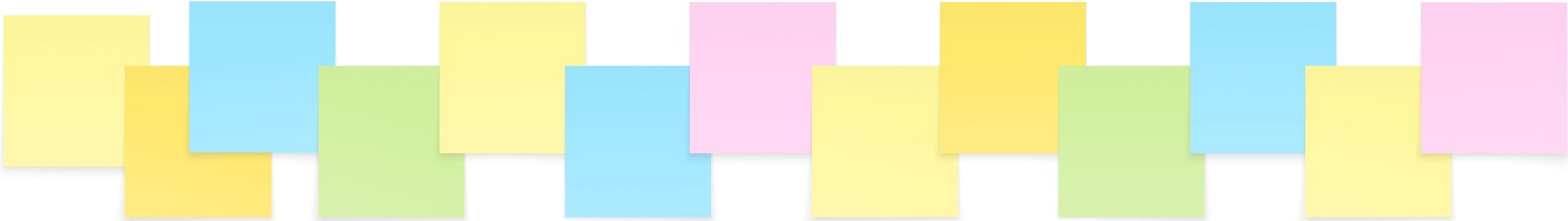
Business Plan 1: Small Business Owner

- Revenue Model: Subscription-based pricing.
- Costs: Development, marketing, support.
- Profitability: Break-even point in 12 months.
- Growth Strategy: Expand to new markets and add new features.

	Column	Column1	Column 2	Column 3	
1	Director Table				
2	Database				
3	Director				
4	PK	Director_ID	Int	not Null	
5		Director Name	NVarChar		
6					
7					
8					
+					



	Column	Column1	Column 2	Column 3	+
1	Actors Table				
2	Database				
3	Actors				
4	Key	Attributes	Data Types	Constraints/comments	
5	PK	Actor_ID	int	not null	
6		Actor Name	nvarchar	not null	
7					
8					
+					

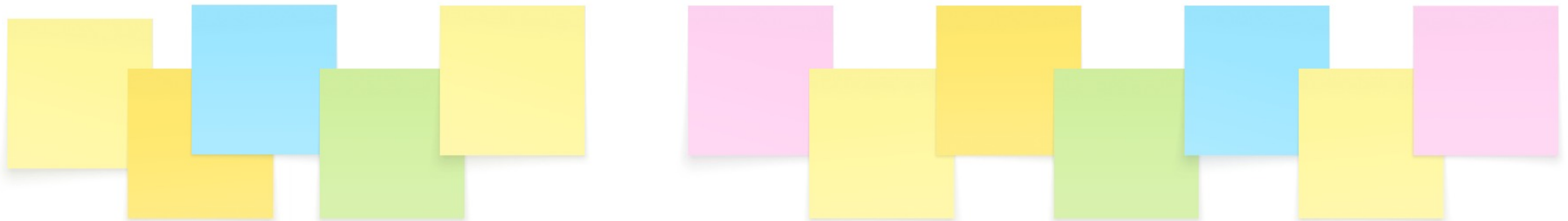


Task 1

Identify and name potential entities from the information supplied.

There are at least 5 entities needed and 2 of these are bridging tables.

1. Movie
2. Director
3. Actor
4. MovieDirector
5. Character

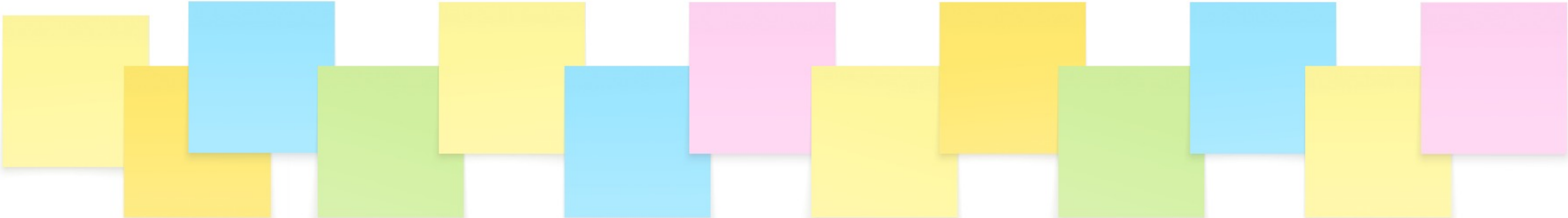


Movies and Actors Database

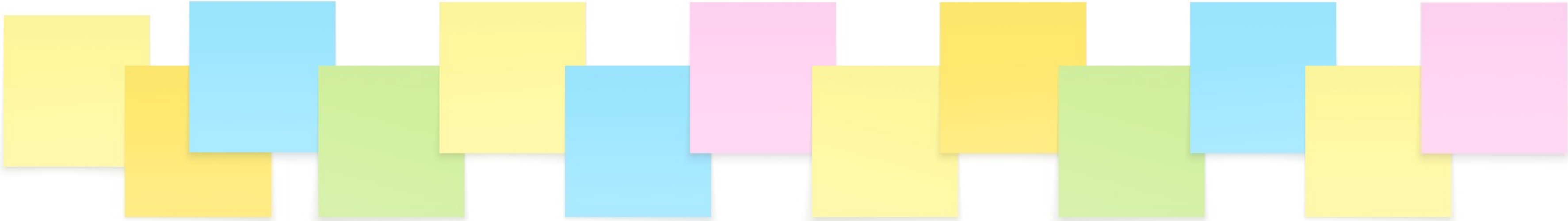
Director	Actor	gender	character_name	cast_order	movie_id	title	budget	homepage	overview	popularity	release_date	revenue	runtime	movie_status	tagline	vote_average	vote_count
Quentin Tarantino	Tim Roth	Male	Ted the Bellhop	43	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hands	8	651
Robert Rodriguez	Tim Roth	Male	Ted the Bellhop	44	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hands	8	651
Allison Anders	Tim Roth	Male	Ted the Bellhop	45	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hands	8	651
Alexandre Rockwell	Tim Roth	Male	Ted the Bellhop	46	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hands	8	651
Quentin Tarantino	Antonio Banderas	Male	Man	47	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hands	8	651
Robert Rodriguez	Antonio Banderas	Male	Man	48	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hands	8	651
Allison Anders	Antonio Banderas	Male	Man	49	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hands	8	651
Alexandre Rockwell	Antonio Banderas	Male	Man	50	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hands	8	651
Quentin Tarantino	Jennifer Beals	Female	Angela	51	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hands	8	651
Robert Rodriguez	Jennifer Beals	Female	Angela	52	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hands	8	651
Allison Anders	Jennifer Beals	Female	Angela	53	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hands	8	651
				54	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hands	8	651

The attributes in the Movies and Actors table are:

- Director
- Actor
- gender
- character_name
- cast_order
- movie_id
- title
- budget
- homepage
- overview
- popularity
- release_date
- revenue
- runtime
- movie_status
- tagline
- vote_average
- vote_count

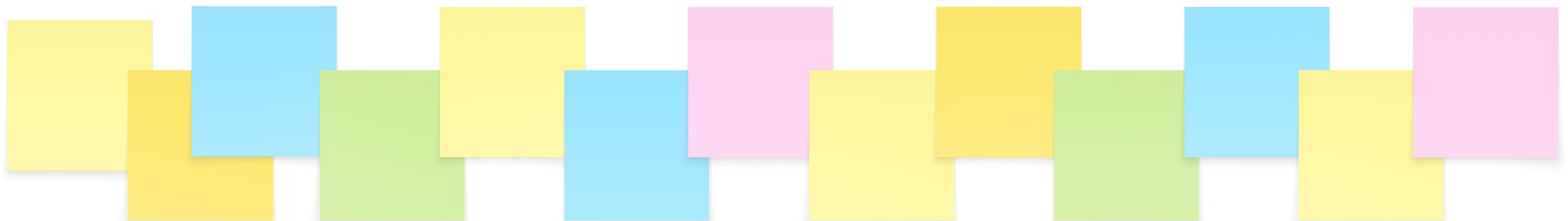
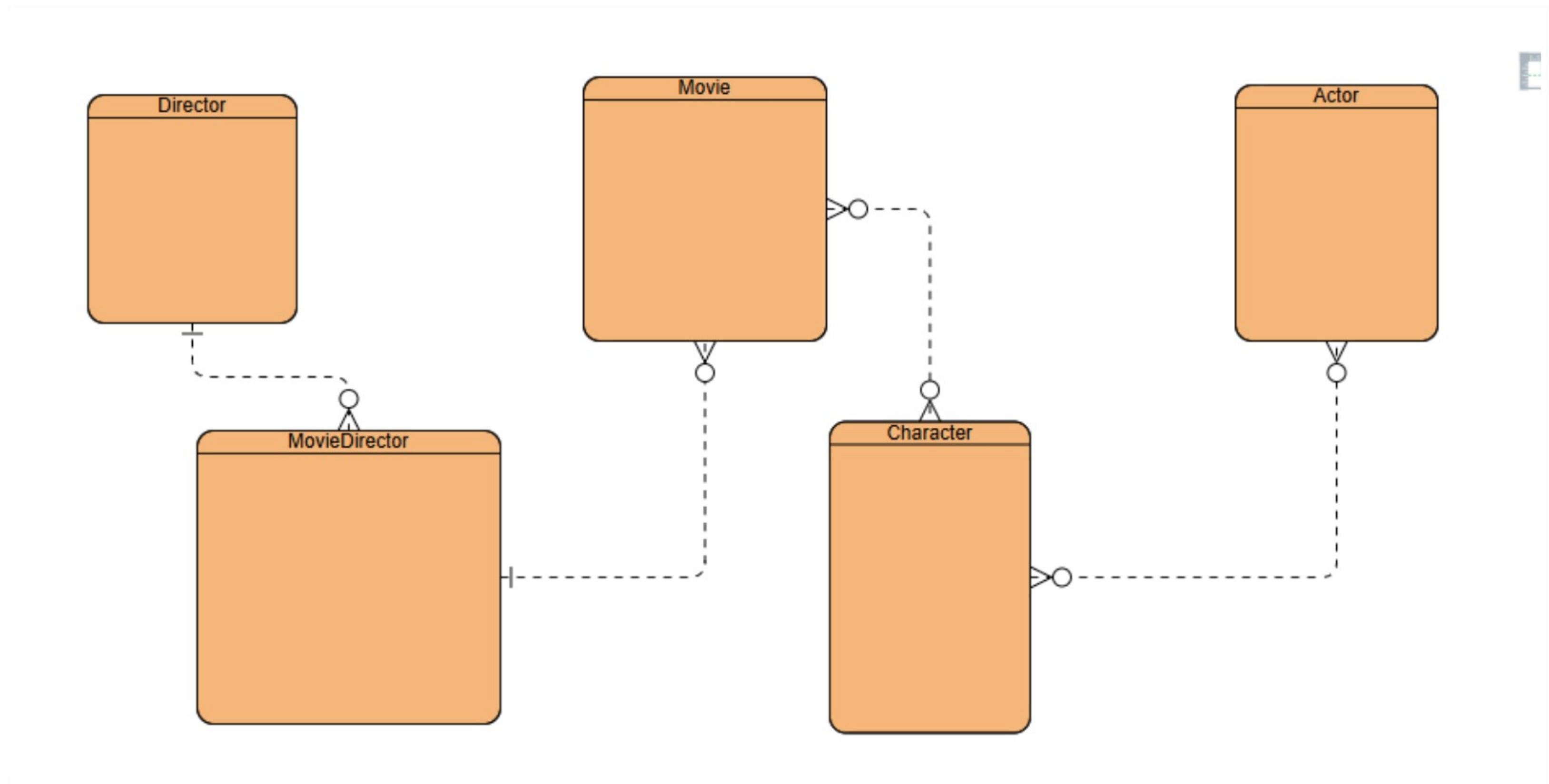


	Column	Column1	Column 2	Column 3	+
1	Character Name				
2	Database				
3	Character				
4	Key	Attributes	Data Types	Constraints/comments	
5	FK	Actor_ID	int	not null	
6	FK	Character_Name	nvarchar	not null	
7		cast order	int	null	
8					
+					

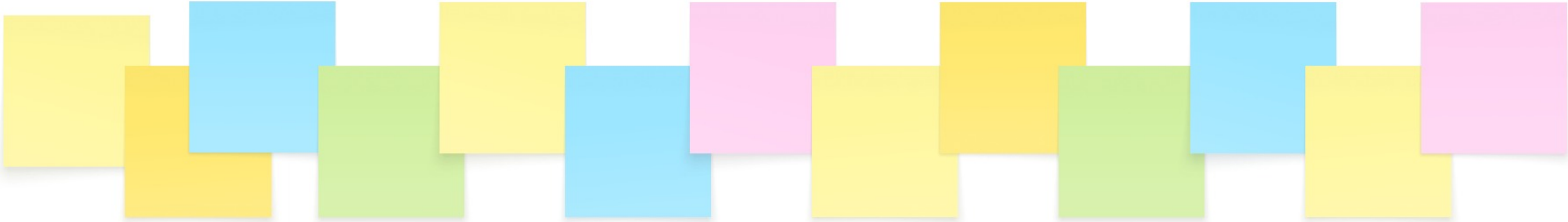


Task 3

Identify entity relationships. Draw an **Entity Relationship Diagram** showing the relationships (cardinality) between the entities (tables) you identified in Task 1.



	Column	Column1	Column 2	Column 3	+
1	MovieDirector				
2	Database				
3	Movie Director				
4	Key	Attributes	Data Types	Constraints/comments	
5	FK	movie id	int	not null	
6	FK	director id	int	not null	
7					
8					
+					



Task 2

Compile data dictionaries for your identified entities.
You may use the pro forma on the following page (or any other valid format).
You should include any Keys (Primary and Foreign), Data Types for each attribute and any suitable constraints I.e. Null, Not Null

	Column	Column1	Column 2	Column 3	+
1	Movies Table				
2	Database				
3	Movies				
4	Key	Attributes	Data Types	Constraints/comments	
5	PK	Movie_id	Int	not null	
6	FK	director_id	Int	not null	
7		Title	nvarchar(100)	not null	
8		budget	money	null	
9		homepage	nvarchar(max)	null	
10		overview	nvarchar(max)	null	
11		popularity	float	null	
12		release_date	date	null	
13		revenue	money	null	
14		runtime	int	null	
15		movie status	nvarchar(50)	not null	
16					
17					



Scenario: A company is planning to launch a new product line. The product line is expected to be profitable in the long run, but the initial investment is high. The company is considering two options: Option A: Launch the product line immediately. Option B: Delay the launch until the market is more mature.

Task 1: Identify the key risks associated with each option.

Option A risks:

- High initial investment
- Market uncertainty
- Competition
- Low demand
- High production costs

Option B risks:

- Delayed market entry
- Increased competition
- Higher production costs
- Loss of market share

Task 2: Evaluate the risks identified in Task 1. Use a risk matrix to assess the likelihood and impact of each risk.

Risk Matrix:

Risk	Likelihood	Impact	Overall Risk
High initial investment	High	High	High
Market uncertainty	Medium	Medium	Medium
Competition	Low	Low	Low
Low demand	Low	High	Medium
High production costs	Low	Medium	Low

Task 3: Develop a risk mitigation strategy for each risk identified in Task 1.

Risk Mitigation Strategy:

Risk	Mitigation Strategy
High initial investment	Conduct a thorough market research before launching the product line.
Market uncertainty	Launch the product line in a limited market to test the demand.
Competition	Develop a competitive advantage through innovation and customer service.
Low demand	Offer discounts and promotions to attract customers.
High production costs	Source raw materials from reliable suppliers and negotiate better prices.

Task 4: Implement the risk mitigation strategy developed in Task 3.

Implementation Plan:

Task	Responsible Person	Start Date	End Date	Status
Conduct market research	John Doe	2023-01-01	2023-01-15	Completed
Launch in limited market	Jane Smith	2023-01-16	2023-01-31	In Progress
Develop competitive advantage	Mike Johnson	2023-01-01	2023-02-15	Not Started
Offer discounts and promotions	Sarah Brown	2023-01-16	2023-01-31	Not Started
Source raw materials	David White	2023-01-01	2023-01-15	Completed

Task 5: Monitor the risks identified in Task 1 and the implementation progress of the risk mitigation strategy.

Risk Monitoring:

Risk	Current Status	Next Steps
High initial investment	Completed	None
Market uncertainty	In Progress	Continue to monitor market response.
Competition	Not Started	Develop competitive advantage.
Low demand	Not Started	Offer discounts and promotions.
High production costs	Completed	None

Task 6: Review the overall performance of the product line launch and the effectiveness of the risk mitigation strategy.

Performance Review:

Metric	Target	Actual	Variance
Revenue	\$1,000,000	\$800,000	-\$200,000
Profit	\$200,000	\$150,000	-\$50,000
Customer Satisfaction	4.5	4.2	-0.3
Market Share	10%	8%	-2%

Task 7: Identify the key lessons learned from the product line launch and the risk mitigation strategy.

Lessons Learned:

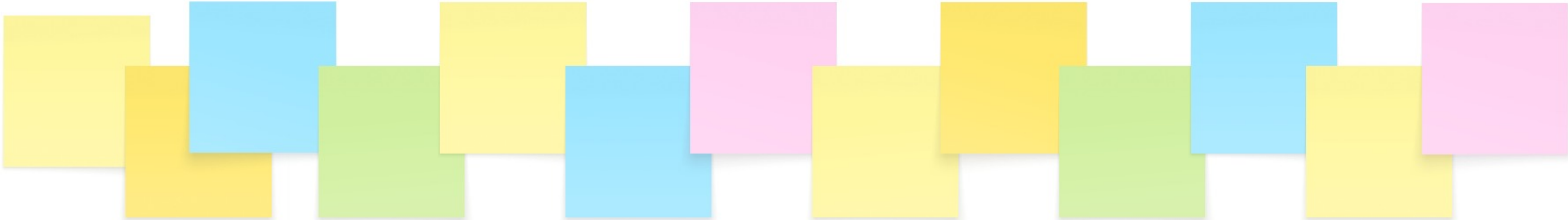
- Thorough market research is essential for product launch.
- Launching in a limited market helps to test the demand.
- Developing a competitive advantage is crucial for long-term success.
- Offering discounts and promotions can help to attract customers.
- Source raw materials from reliable suppliers to ensure quality and cost control.

Task 8: Develop a risk management plan for the company's future product line launches.

Risk Management Plan:

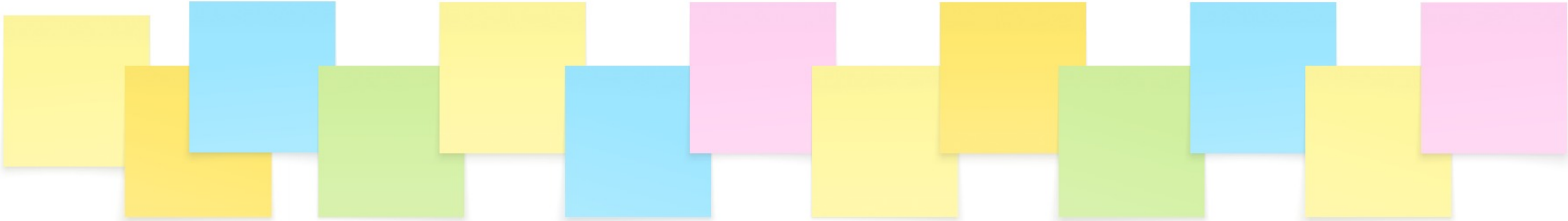
Risk	Identification	Assessment	Mitigation	Monitoring
High initial investment	Identify the investment amount and the expected return.	Assess the investment risk using a risk matrix.	Conduct a thorough market research before launching the product line.	Monitor the investment progress and the market response.
Market uncertainty	Identify the market trends and the potential competition.	Assess the market uncertainty risk using a risk matrix.	Launch the product line in a limited market to test the demand.	Continue to monitor market response.
Competition	Identify the competitive advantage and the potential competitors.	Assess the competition risk using a risk matrix.	Develop a competitive advantage through innovation and customer service.	Monitor the competitive advantage and the market share.
Low demand	Identify the potential demand and the potential barriers.	Assess the low demand risk using a risk matrix.	Offer discounts and promotions to attract customers.	Monitor the customer satisfaction and the market share.
High production costs	Identify the raw materials and the production process.	Assess the high production costs risk using a risk matrix.	Source raw materials from reliable suppliers and negotiate better prices.	Monitor the production costs and the quality control.

	Column	Column1	Column 2	Column 3	+
1	Entity description	Overview of movies			
2	Database	Movies			
3	Entity name	Movie			
4	Key	Attributes	Data Types	Constraints/comments	
5	PK	Movie ID	int	not null	
6		title	VCHAR(100)	not null	
7		budget	money	null	
8		homepage	VCHAR(300)	null	
9		overview	VCHAR(300)	null	
10		popularity	int	null	
11		release date	date	not null	
12		revenue	money	null	
13		runtime	int	not null	
14		movie status	VARCHAR(50)	not null	
15		Tagline	VCHAR(300)	null	
16		Vote_Average	int	null	
17		Vote_Count	int	null	
+					



Entity 3

	Column	Column1	Column 2	Column 3	
1	Entity description	Actor			
2	Database	Movies			
3	Entity name	Actor			
4	Key	Attributes	Data Types	Constraints/comments	
5	PK	Actor ID	INT	Not Null	
6		Actor Name	VARCHAR (100)	Not Null	
7		Gender	VARCHAR (10)	Not Null	
8					
+					

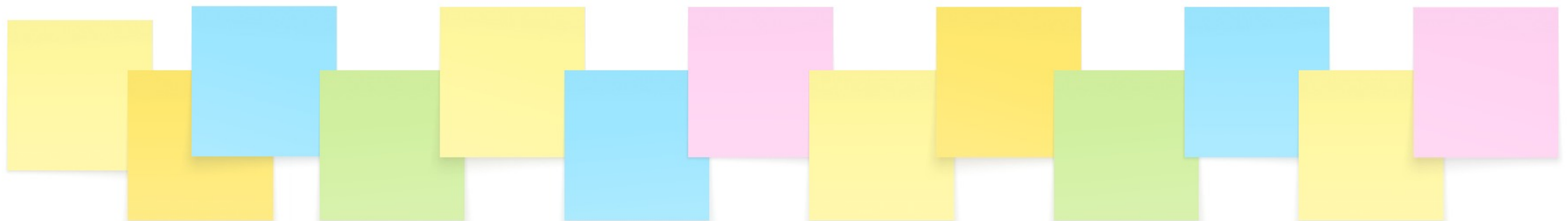


Task 1

Identify and name potential entities from the information supplied.

There are at least 5 entities needed and 2 of these are bridging tables.

1. Director
2. Movie
3. Actor
4. Character (Bridging)
5. Movie and Director (Bridging)

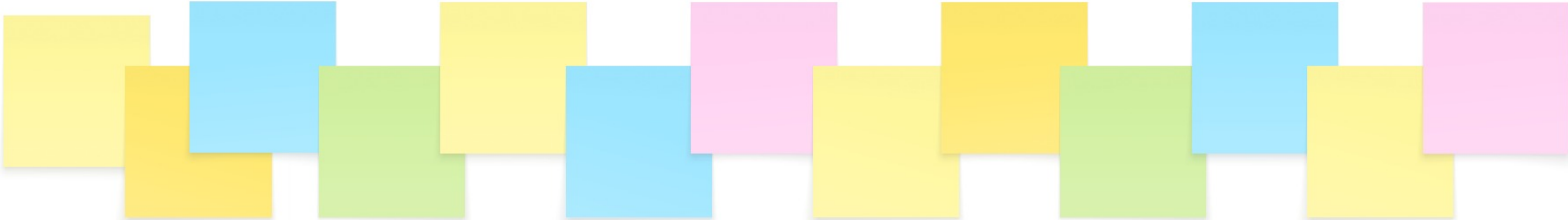


Movies and Actors Database

Director	Actor	gender	character_name	cast_order	movie_id	title	budget	homepage	overview	popularity	release_date	revenue	runtime	movie_status	tagline	vote_average	vote_count
Quentin Tarantino	Tim Roth	Male	Ted the Bellhop	43	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hands	8	651
Robert Rodriguez	Tim Roth	Male	Ted the Bellhop	44	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hands	8	651
Allison Anders	Tim Roth	Male	Ted the Bellhop	45	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hands	8	651
Alexandre Rockwell	Tim Roth	Male	Ted the Bellhop	46	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hands	8	651
Quentin Tarantino	Antonio Banderas	Male	Man	47	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hands	8	651
Robert Rodriguez	Antonio Banderas	Male	Man	48	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hands	8	651
Allison Anders	Antonio Banderas	Male	Man	49	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hands	8	651
Alexandre Rockwell	Antonio Banderas	Male	Man	50	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hands	8	651
Quentin Tarantino	Jennifer Beals	Female	Angela	51	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hands	8	651
Robert Rodriguez	Jennifer Beals	Female	Angela	52	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hands	8	651
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				54	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hands	8	651

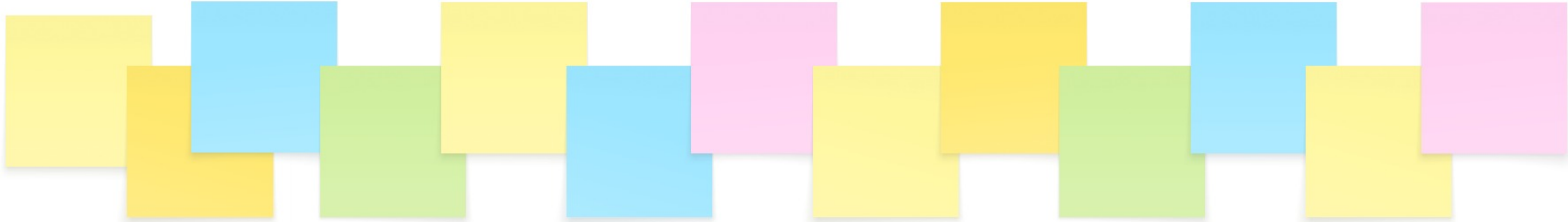
The attributes in the Movies and Actors table are:

- Director 1
- Actor 3
- gender 3
- character_name 3
- cast_order 3
- movie_id 2
- title 2
- budget 2
- homepage 2
- overview 2
- popularity 2
- release_date 2
- revenue 2
- runtime 2
- movie_status 2
- tagline 2
- vote_average 2
- vote_count 2



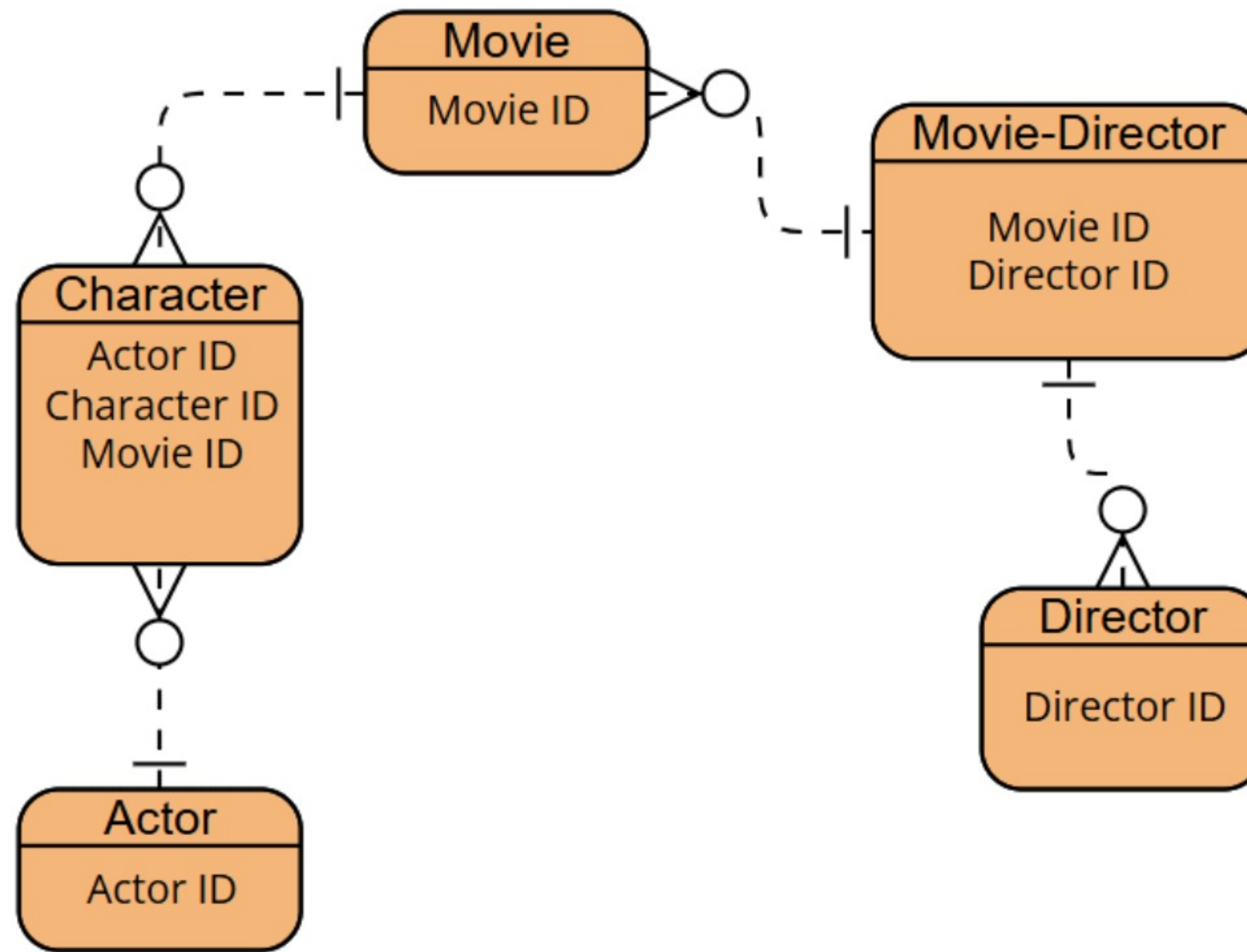
Entity 5

	Column	Column1	Column 2	Column 3	+
1	Entity description	Movie-Director			
2	Database	Movies			
3	Entity name	Movie and Director			
4	Key	Attributes	Data Types	Constraints/comments	
5	FK	Movie ID	INT	NOT NULL	
6	FK	Director ID	INT	NOT NULL	
7					
8					
+					



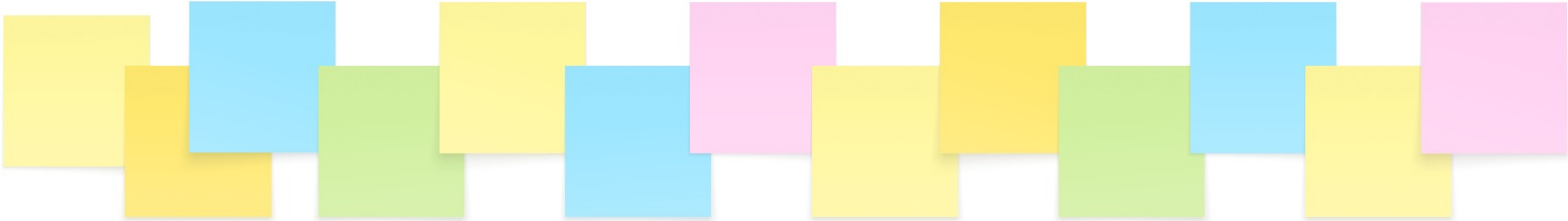
Task 3

Identify entity relationships. Draw an **Entity Relationship Diagram** showing the relationships (cardinality) between the entities (tables) you identified in Task 1.



Entity 4

	Column	Column1	Column 2	Column 3	+
1	Entity description	Character			
2	Database	Movies			
3	Entity name	Character			
4	Key	Attributes	Data Types	Constraints/comments	
5	PK	Character ID	int	NOT NULL	
6	FK	ActorID	INT	NOT NULL	
7	FK	MovieID	INT	NOT NULL	
8		Character_Name	VARCHAR(255)	NULL	
9		Cast_Order	INT	NULL	
+					



Task 2

Compile data dictionaries for your identified entities.
You may use the pro forma on the following page (or any other valid format).
You should include any Keys (Primary and Foreign), Data Types for each attribute and any suitable constraints I.e. Null, Not Null

	↺	📁	📄	▽	↕	🗑	⚙	📄	🔍
	📄	Column	☰	Column1	☰	Column 2	☰	Column 3	+
1		Entity description		Director of movies					
2		Database		Movies					
3		Entity name		Director					
4		Key		Attributes		Data Types		Constraints/comments	
5		PK		Director ID		Int		not null	
6				Director Name		VCHAR(50)		not null	
7									
8									
+									

